

Equation List for Exam

The listed equations serve the purpose of preparation for the calculation part of the aircraft design exam and should be understood and learned by heart. Exam candidates must also be able to derive further relations from these equations.

Chapter 2: Aerodynamics

2.1	Wing geometrical definitions	$AR = \frac{b^2}{S_{ref}}$ $\lambda = \frac{c_{tip}}{c_{root}}$ $c_{MAC} = \frac{2}{3} c_{root} \frac{1 + \lambda + \lambda^2}{1 + \lambda}$
2.1	Lift, drag and moment	$L = \frac{1}{2} \rho V^2 C_L S_{ref}$ $D = \frac{1}{2} \rho V^2 C_D S_{ref}$ $M = \frac{1}{2} \rho V^2 C_M c_{MAC} S_{ref}$
2.1	Lifting line theory, Polhamus method, Slender body theory	Lifting line theory: $C_{L\alpha} = \frac{2 \pi AR}{2 + AR}$ Polhamus method: $C_{L\alpha} = \frac{2 \pi AR}{2 + \sqrt{\frac{AR^2 \beta^2}{K^2} \left(1 + \frac{\tan^2 \Lambda_{50\%}}{\beta^2}\right)} + 4}$ with $\beta = \sqrt{1 - M^2}$ (subsonic) Slender body theory: $C_{L\alpha} = \frac{\pi AR}{2}$
2.2	Components of drag coefficient	$C_{D0} = C_f + C_{D_{fi}} + C_{D_W}$
2.2	Equivalent aircraft & component parasite drag area	$D_0 = \frac{1}{2} \rho V^2 S_{ref} C_{D0} = \frac{1}{2} \rho V^2 \sum_{j=1}^n S_j C_{D0,j}$

2.2	Reynolds number	$Re = \frac{\rho V l}{\mu}$
2.2	Component drag coefficient and parasite drag area	$C_{D0,j} = c_{f,j} F_j$ $f_j = c_{f,j} F_j S_j$
2.2	Lift-induced drag coefficient	With $C_{Di} = k C_L^2$ Or $k = \frac{1}{C_{L\alpha}}$ $k = \frac{1}{\pi AR e}$
2.6	Stability measure	$\frac{\partial C_M}{\partial C_L} = \frac{x_{CG}}{c_{MAC,W}} - \frac{x_{AC}}{c_{MAC,W}} = (\bar{x}_{CG} - \bar{x}_{AC}) [-]$ $= (\bar{x}_{CG} - \bar{x}_{AC}) * 100 [\%]$
2.6	Aerodynamic center of wing or horizontal tail	Or $\bar{x}_{AC,W} = \frac{x_{0,W} + (n_{AC,W} \cdot c_{root,W})}{c_{MAC}}$ $\bar{x}_{AC,HT} = \frac{x_{0,HT} + (n_{AC,HT} \cdot c_{root,HT})}{c_{MAC}}$
2.6	Neutral point calculation	$\bar{x}_{AC} = \frac{C_{M\alpha,W} - C_{M\alpha,fus} + C_{M\alpha,HT}}{C_{L\alpha,W} + C_{L\alpha,HT}} =$ $\frac{C_{L\alpha,W} \bar{x}_{AC,W} - C_{M\alpha,fus} + \left(\frac{q_{HT}}{q} \frac{\partial \alpha_{HT}}{\partial \alpha} \frac{S_{HT}}{S_{ref}} C'_{L\alpha,HT} \right) \bar{x}_{AC,HT}}{C_{L\alpha,W} + \left(\frac{q_{HT}}{q} \frac{\partial \alpha_{HT}}{\partial \alpha} \frac{S_{HT}}{S_{ref}} C'_{L\alpha,HT} \right)}$
2.6	Trim around lateral axis	$C_{M(x=0)} = C_{M(CL=0)} + \frac{\partial C_M}{\partial C_L} C_L$

Chapter 3: Structural design

3.1	Weight breakdown of take-off and landing weight	$w_{TO} = w_{OE} + w_{fuel} + w_{PL}$ $w_L = w_{OE} + w_{PL}$ (w_L assuming no fuel at end of mission)
3.3	Growth factor for payload mass (example)	$\partial m_{TO} = \left(\frac{m_{TO}}{m_{PL}} \right)_{\%} \partial m_{PL}$
3.4	Vertical load factor	$n_z = \frac{L}{w}$
3.4	Load factors during maneuvers	$n_z = \frac{L_0 + \Delta L_{man}}{w}$ $n_z = \frac{w / \cos(\phi)}{w}$
3.4	Conversion from TAS to EAS	$V_{EAS} = \sqrt{\frac{\rho_H}{\rho_0}} V_{TAS}$

Chapter 4: Propulsion design

4.2	Engine thrust	$\frac{T}{T_i} = \left(\frac{V}{V_i} \right)^{n_v} \left(\frac{\rho}{\rho_i} \right)^{n_p}$
4.3	Thrust and power specific fuel consumption	$TSFC = \frac{\dot{m}_{fuel}}{T}$ $PSFC = \frac{\dot{m}_{fuel}}{P}$

Chapter 5: Point performance

5.2	Specific excess power	$SEP = \frac{T \cos(\alpha + \sigma) - D}{m g} V$
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Chapter 6: Mission performance

6.3	Weight reduction rate	$dw = -\dot{m}_{fuel} g dt$
6.3	Flight condition I: duration and range	$\Delta t = \frac{V^{n_V}}{g SFC_x} \left(\frac{C_L}{C_D} \right)_x \ln \frac{w_i}{w_{i+1}}$ during flight phase x $\Delta s = \frac{V^{n_V+1}}{g SFC_x} \left(\frac{C_L}{C_D} \right)_x \ln \frac{w_i}{w_{i+1}}$ during flight phase x Turbojet/turbofan: $SFC_x = TSFC_x; n_V \approx 0$ Turboprop: $SFC_x = PSFC_x; n_V \approx -1$

Chapter 7: Design chart

7.1	Definition of thrust-to-weight ratio	$\frac{T}{w} = \frac{Thrust}{Weight} [-]$
7.1	Definition of wing loading	$\frac{w}{S} = \frac{Weight}{Surface} \left[\frac{N}{m^2} \right]$

General

	Lift coefficient, lift curve slope and angle of attack	$C_L = C_{L(\alpha=0^\circ)} + C_{L\alpha} \alpha$
	Mach number	$M = \frac{V}{a}$